

Sharpe Ratio and Risk & Risk Premiums (BKM 13e §5.3)

FIN 4000 Investments — General Study Guide gen-10 (Chapter 5)

Big picture: §5.3 turns “how did the investment do?” into “how much reward did I earn per unit of risk?” You compute expected return and standard deviation from scenarios, subtract the risk-free rate to get the risk premium, and divide by σ to get the Sharpe ratio, the number used to compare portfolios on equal footing.

Learning Objectives

By the end of this guide you should be able to:

1. Compute a holding-period return (HPR) from prices and dividends.
 2. Compute **expected return**, **variance**, and **standard deviation** from a scenario table.
 3. Distinguish **risk premium** (expected) from **excess return** (realized).
 4. Compute and interpret the **Sharpe ratio** (reward-to-volatility ratio) from scenario data or from historical excess returns.
 5. Compare portfolios using the Sharpe ratio, and explain why mixing a portfolio with T-bills does not change it.
 6. Annualize a monthly Sharpe ratio.
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Core Concept

1. Holding-period return

$$HPR = \frac{P_1 - P_0 + D_1}{P_0}$$

Return is uncertain, so we describe it with a **scenario analysis**: a list of states s , each with a probability $p(s)$ and a return $r(s)$.

2. Expected return, variance, standard deviation

$$E(r) = \sum_s p(s) r(s)$$
$$\sigma^2 = \sum_s p(s) [r(s) - E(r)]^2$$
$$\sigma = \sqrt{\sigma^2}$$

σ is the standard measure of total risk in this chapter.

3. Risk-free rate, risk premium, excess return

- **Risk-free rate** (r_f): the return on T-bills, the benchmark safe asset.
- **Risk premium** (what you *expect* to earn for bearing risk):

$$\text{Risk premium} = E(r) - r_f$$

- **Excess return** (what you *actually* earned in a given period):

$$\text{Excess return} = r - r_f$$

The risk premium is the expected value of the excess return.

4. The Sharpe ratio

$$\text{Sharpe ratio} = \frac{E(r_P) - r_f}{\sigma_P}$$

- The numerator is the risk premium; the denominator is total risk. - Interpretation: extra return earned per 1% of standard deviation. Higher is better. - With historical data, let $R_t = r_t - r_{f,t}$ be the excess return in period t . Use the sample mean $\bar{R}$ and sample standard deviation s_R of those excess returns:

$$\text{Sharpe ratio} = \frac{\bar{R}}{s_R}$$

- The Sharpe ratio is the **slope of the capital allocation line (CAL)** formed by combining the risk-free asset with portfolio P . - Because every mix of P and T-bills lies on the same line, **blending with T-bills (or borrowing at the risk-free rate) changes risk and return but not the Sharpe ratio.**

5. Annualizing

If the monthly Sharpe ratio is S_m , then

$$S_{\text{annual}} \approx S_m \times \sqrt{12}$$

The mean scales with 12, but σ scales with $\sqrt{12}$.

Worked Examples

Example 1: Scenario analysis and Sharpe ratio

A stock has these scenarios. The T-bill rate is $r_f = 3\%$.

State	Probability	Return
Boom	0.30	30%
Normal	0.50	12%
Recession	0.20	-18%

Expected return:

$$E(r) = 0.30(30) + 0.50(12) + 0.20(-18) = 9.0 + 6.0 - 3.6 = 11.4\%$$

Variance:

$$\begin{aligned}\sigma^2 &= 0.30(30 - 11.4)^2 + 0.50(12 - 11.4)^2 + 0.20(-18 - 11.4)^2 \\ \sigma^2 &= 0.30(345.96) + 0.50(0.36) + 0.20(864.36) = 103.788 + 0.18 + 172.872 = 276.84\end{aligned}$$

Standard deviation:

$$\sigma = \sqrt{276.84} = 16.64\%$$

Risk premium:

$$E(r) - r_f = 11.4 - 3 = 8.4\%$$

Sharpe ratio:

$$\text{Sharpe ratio} = \frac{8.4}{16.64} = 0.505$$

BA II Plus (statistics worksheet, probabilities entered as frequencies out of 100):

1. 2nd DATA (opens the data worksheet), then 2nd CLR WORK
2. X01 = 30 ENTER ↓, Y01 = 30 ENTER ↓
3. X02 = 12 ENTER ↓, Y02 = 50 ENTER ↓
4. X03 = 18 +/- ENTER ↓, Y03 = 20 ENTER
5. 2nd STAT, press 2nd ENTER until the mode reads LIN, then ↓ through the results:
 - $\bar{x} = 11.4$ (this is $E(r)$)
 - $\sigma_x = 16.64$ (**use the population value**, not the sample S_x , because the probabilities are the whole distribution)

Then compute the Sharpe ratio by hand: $(11.4 - 3) \div 16.64$.

Example 2: Comparing two portfolios

$$r_f = 3\%.$$

Portfolio	$E(r)$	σ
A	12%	20%
B	9%	11%

$$\text{Sharpe}_A = \frac{12 - 3}{20} = 0.450 \quad \text{Sharpe}_B = \frac{9 - 3}{11} = 0.545$$

B is the better risk-adjusted performer even though A has the higher expected return. Per unit of risk, B pays more.

Example 3: Sharpe ratio from historical excess returns

Five years of a fund's **excess returns** $R_t = r_t - r_{f,t}$: 10%, -8%, 18%, 6%, 4%.

$$\bar{R} = \frac{10 - 8 + 18 + 6 + 4}{5} = 6.0\%$$

Deviations from the mean: 4, -14, 12, 0, -2. Squared: 16, 196, 144, 0, 4. Sum = 360.

$$s^2 = \frac{360}{5 - 1} = 90 \quad s = \sqrt{90} = 9.49\%$$

$$\text{Sharpe ratio} = \frac{6.0}{9.49} = 0.632$$

BA II Plus:

1. 2nd DATA, 2nd CLR WORK
 2. Enter X01 = 10, X02 = 8 +/-, X03 = 18, X04 = 6, X05 = 4 (press ENTER ↓ after each)
 3. 2nd STAT (LIN mode), ↓ to read $\bar{x} = 6.0$ and $S_x = 9.49$ (**use the sample value** for historical data)
 4. $6.0 \div 9.49 = 0.632$
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Example 4: Mixing a risky portfolio with T-bills

Portfolio P has $E(r_P) = 14\%$, $\sigma_P = 22\%$, and $r_f = 4\%$.

$$\text{Sharpe}_P = \frac{14 - 4}{22} = 0.4545$$

Target standard deviation of 11% (invest weight y in P , the rest in T-bills):

$$y = \frac{11}{22} = 0.50 \quad E(r) = r_f + y[E(r_P) - r_f] = 4 + 0.50(10) = 9\%$$

$$\text{Sharpe} = \frac{9 - 4}{11} = 0.4545 \quad (\text{unchanged})$$

Target standard deviation of 33% (borrow at r_f to invest $y = 1.5$ in P):

$$E(r) = 4 + 1.5(10) = 19\% \quad \text{Sharpe} = \frac{19 - 4}{33} = 0.4545 \quad (\text{unchanged})$$

Leverage and de-leveraging slide you along the same CAL; the slope is the Sharpe ratio.

Example 5: Annualizing a monthly Sharpe ratio

A fund has an average monthly excess return of 0.6% and a monthly standard deviation of 3.0%.

$$S_m = \frac{0.6}{3.0} = 0.20 \quad S_{\text{annual}} = 0.20 \times \sqrt{12} = 0.693$$

Check: annual mean excess return = $0.6 \times 12 = 7.2\%$; annual $\sigma = 3.0 \times \sqrt{12} = 10.39\%$; $7.2 \div 10.39 = 0.693$.

Common Pitfalls

- **Forgetting to subtract the risk-free rate.** Return $\div \sigma$ is *not* the Sharpe ratio. The numerator must be the excess return or risk premium.
- **Population vs. sample standard deviation.** Scenario tables (the full probability distribution) use the population σ_x . A time series of historical returns uses the sample S_x (divide by $n - 1$).
- **Mixing units.** Keep $E(r)$, r_f , and σ all in percent (or all in decimals) before dividing.

- **Risk premium vs. excess return.** The risk premium is *expected*; the excess return is *realized*. Scenario problems use the premium; historical problems use average excess returns.
- **Comparing Sharpe ratios built on different risk-free rates or different periods.** Compare like with like.
- **Thinking leverage changes the Sharpe ratio.** Mixing with T-bills or borrowing at r_f scales the premium and σ by the same factor, so the ratio is unchanged.
- **Annualizing the standard deviation by 12.** σ scales with $\sqrt{12}$, not 12, and so does the Sharpe ratio.
- **Misreading a negative Sharpe ratio.** A negative value means the portfolio earned less than T-bills. Interpret the sign before ranking.
- **Treating standard deviation as the only risk.** The Sharpe ratio uses *total* risk, which suits an investor's whole portfolio; it is not a beta-based measure.

Key Terms Checklist

Make sure you can define each in your own words.

☐ holding-period return (HPR)	☐ scenario analysis	☐ probability distribution
☐ expected return $E(r)$	☐ variance σ^2	☐ standard deviation σ
☐ risk-free rate r_f	☐ risk premium	☐ excess return
☐ reward-to-volatility (Sharpe) ratio	☐ capital allocation line (CAL)	☐ risk aversion
☐ population vs. sample standard deviation	☐ annualizing ($\sqrt{12}$ rule)	☐ total risk

Practice Questions (CFA-style, choose A, B, or C)

Q1. The Sharpe ratio measures:

- A. excess return per unit of total risk (standard deviation)
- B. excess return per unit of beta
- C. total return per unit of standard deviation

Q2. A portfolio has an expected return of 10%, a standard deviation of 15%, and the T-bill rate is 2.5%. The Sharpe ratio is closest to:

- A. 0.50
- B. 0.67
- C. 0.17

Q3. Fund X has $E(r) = 11\%$ and $\sigma = 18\%$. Fund Y has $E(r) = 8\%$ and $\sigma = 10\%$. The risk-free rate is 3%. Based on the Sharpe ratio:

- A. Fund X is better because its expected return is higher
- B. Fund Y is better
- C. The funds are equivalent

Q4. A stock has three scenarios: 25% probability of a 40% return, 50% probability of a 10% return, and 25% probability of a -20% return. The standard deviation of returns is closest to:

- A. 21.2%

- B. 15.0%
- C. 450%

Q5. An investor moves part of a risky portfolio P into T-bills to reduce risk. The Sharpe ratio of the resulting mix will:

- A. increase
- B. stay the same
- C. decrease

Q6. A fund's monthly Sharpe ratio is 0.25. The annualized Sharpe ratio is closest to:

- A. 0.87
- B. 3.00
- C. 0.07

Q7. The slope of the capital allocation line combining the risk-free asset with risky portfolio P equals:

- A. the beta of P
- B. the Sharpe ratio of P
- C. the expected return of P

Q8. Over three years, a fund's excess returns were 12%, -4%, and 10%. Using the sample standard deviation, the Sharpe ratio is closest to:

- A. 0.69
- B. 0.84
- C. 0.50

Answer Key

Q1. A. Sharpe = $[E(r) - r_f] \div \sigma$. B describes a Treynor-style measure (per unit of beta), and C leaves out the subtraction of r_f .

Q2. A.

$$\frac{10 - 2.5}{15} = \frac{7.5}{15} = 0.50$$

B (0.67) divides by 11.25, which is not the standard deviation. C (0.17) divides by the wrong quantity.

Q3. B.

$$\text{Sharpe}_X = \frac{11 - 3}{18} = 0.444 \quad \text{Sharpe}_Y = \frac{8 - 3}{10} = 0.500$$

Y earns more premium per unit of risk. A ignores risk.

Q4. A.

$$\begin{aligned} E(r) &= 0.25(40) + 0.50(10) + 0.25(-20) = 10\% \\ \sigma^2 &= 0.25(30)^2 + 0.50(0)^2 + 0.25(-30)^2 = 225 + 0 + 225 = 450 \\ \sigma &= \sqrt{450} = 21.2\% \end{aligned}$$

C reports the variance, not the standard deviation. B is not supported by any calculation.

Q5. B. Mixing P with T-bills moves you along the same CAL, scaling the premium and σ by the same factor, so the slope (Sharpe ratio) does not change.

Q6. A.

$$S_{\text{annual}} = 0.25 \times \sqrt{12} = 0.25 \times 3.464 = 0.866 \approx 0.87$$

B multiplies by 12 (wrong scaling). C divides by $\sqrt{12}$ (wrong direction).

Q7. B. The CAL runs from $(0, r_f)$ through P ; its slope is $[E(r_P) - r_f] \div \sigma_P$, the Sharpe ratio.

Q8. A.

$$\bar{R} = \frac{12 - 4 + 10}{3} = 6.0\%$$

Deviations: 6, -10, 4. Squared: 36, 100, 16. Sum = 152.

$$s^2 = \frac{152}{2} = 76 \quad s = 8.72\% \quad \text{Sharpe ratio} = \frac{6.0}{8.72} = 0.69$$

B uses the population σ (7.12%), which is wrong for historical sample data.

Study guide gen-10 for BKM (13th ed.), Chapter 5, §5.3 "Risk and Risk Premiums" (Sharpe ratio / reward-to-volatility ratio). All numeric answers were checked with Python.